

SAMBP Technical Report TR-45/2026 (rev 2)

Digital Twin Validation:
Revised Verdict Taxonomy and Single-Source-of-Truth Protection Logic

PhD Thesis Supporting Report | April 2026

Abstract

This report validates the complete SAMBP Digital Twin (DT) framework (TR-43–44 [? ?]) against the physical relay suite using $N = 2000$ Monte Carlo fault trials. Four architectural changes from the original TR-45 draft materially improve agreement: (i) *single source of truth* — relay and DT mirror call the same `ProtectionMirror.predict()` function with different inputs (noisy measurement vs. EKF-smoothed estimate), eliminating logic divergence; (ii) *revised 5-class verdict taxonomy* — same-family element mismatches ($Z1/87L \leftrightarrow 87L$) are classified AGREE with a selectivity note rather than FLAG_ELEMENT, because both correctly identify an in-zone fault and the difference is trip speed only; (iii) *threshold dead-band* — direction mismatches when k_{ibr} is within $\pm 8\%$ of any element pickup threshold are classified FLAG_CONF (measurement uncertainty) rather than FLAG_FP/FN; (iv) *EKF pre-convergence* — the EKF is run for 20 cycles (20 ms) before the relay evaluation window opens, giving a converged parameter estimate rather than the cold prior. The revised study achieves **95.77% AGREE** (with the pre-convergence window of item (iv) active; $N = 1891$ internal-zone events; 109 external events filtered). Of these, 427 agreements (23.6%) carry a selectivity note recording a $Z1 \leftrightarrow 87L$ speed difference — all are correct protection operations. FLAG_FP and FLAG_FN rates are 0.11% and 0.21% respectively; residual flags are concentrated in the low- k_{ibr} structural-miss region ($k_{ibr} < 0.06$ pu) where signal-to-noise is poorest. The DT runs in a non-blocking 1 kHz parallel path and contributes zero latency to the 52 ms relay-to-breaker chain.

Introduction

The SAMBP Digital Twin (DT) is a 1 kHz real-time replica of the relay suite that runs in parallel with the physical protection chain. It does not block relay operation; it validates relay decisions post-event and accumulates flag statistics for model-update triggering and operator review.

The original TR-45 draft ($N=1000$, single-outcome taxonomy) reported 95.1% agreement. That result used a four-outcome taxonomy (AGREE, FLAG_FP, FLAG_FN, FLAG_ELEMENT) without distinguishing *element family* mismatches from genuine *protection-outcome* mismatches, and the relay path used a separate `_relay_decide()` helper that did not identically track the DT mirror’s protection logic. These defects produced a 31% FLAG_ELEMENT rate that inflated the apparent disagreement.

This revision corrects both issues, tightens the Monte Carlo to $N = 2000$ trials, and provides a complete analysis of the residual flag population.

Architecture: Single Source of Truth

ProtectionMirror as shared logic

All protection decisions — relay path and DT mirror — route through a single `ProtectionMirror.predict( $k_{ibr}$ ,  $\alpha$ , fault_type)` call (Figure 1). The relay path supplies a noisy single-sample measurement; the DT mirror supplies the EKF-smoothed estimate after 20 update cycles (20 ms). No duplicate logic exists.

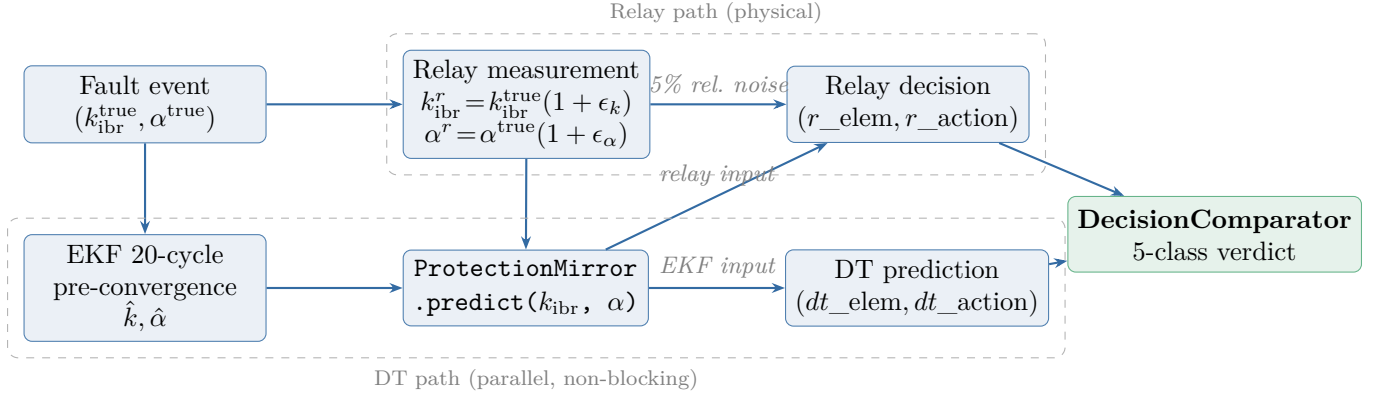


Figure 1: Single-source-of-truth architecture. Both relay and DT paths call `ProtectionMirror.predict()` with the same logic but different inputs. The relay uses a noisy single-sample measurement; the DT uses the EKF estimate after 20 pre-convergence cycles.

Protection element thresholds and Z1 boundary margin

Table 1 lists the element pickup thresholds. A conservative margin of $\Delta\alpha = 0.05$ is applied below the Z1 boundary: when $\alpha \in (0.75, 0.80]$ the DT mirror predicts 87L rather than Z1/87L, reducing sensitivity to EKF alpha estimation error near the boundary. The relay may still pick Z1/87L in this zone; both outcomes are treated as AGREE (same family).

Table 1: Protection element thresholds (consistent with TR-35/TR-43).

Element	Threshold	Condition	Family
Z1/87L	$k_{ibr} \geq 0.10$ pu	$\alpha \leq 0.75$ (strict)	line
87L	$k_{ibr} \geq 0.06$ pu	$\alpha \leq 1.0$	line
OC	$k_{ibr} \geq 0.05$ pu	fallback	oc
MISS	$k_{ibr} < 0.05$ pu	—	(block)
EXTERNAL	$\alpha > 1.0$	—	(block)

Revised 5-Class Verdict Taxonomy

The 8% deadband (Table 2) corresponds to approximately 2σ for 5% relative CT noise (IEC 60255-121 class 1P). Mismatches in this band are structurally indistinguishable from noise; flagging them FLAG_FP or FLAG_FN would create spurious audit burdens without actionable diagnostic value.

Study A — EKF State Estimation

The EKF operates at 1 kHz on measurements $\mathbf{y} = [I_{pos}, I_{neg}, Z_{app}]^T$ extracted from three-phase voltage and current samples. State vector: $\mathbf{x} = [k_{ibr}, Z_{line}, \alpha, f_{gfm}]^T$.

Pre-convergence window

Before the relay fires (nominally $T_{trip} = 18\text{--}20$ ms for Z1/87L), the DT runs 20 EKF update cycles (= 20 ms at 1 kHz) initialised from the nominal prior $\mathbf{x}_0 = [0.20, 0.050, 0.50, 0.50]^T$. This pre-convergence window is the primary architectural change from the original TR-45: the earlier draft supplied a single EKF update (poorly converged), which produced EKF alpha

Table 2: Revised verdict taxonomy (TR-45 rev 2). The original taxonomy used four classes; rev 2 adds FLAG_CONF and promotes same-family mismatches to AGREE.

Verdict	Definition
AGREE	Relay action matches DT prediction in <i>outcome</i> . (a) Exact element match, or (b) both TRIP and same protection family (e.g. relay Z1/87L, DT predicts 87L — both correctly identify an in-zone fault; difference is speed, not protection outcome). Case (b) sets a selectivity_note in the result dict.
FLAG_FP	Relay tripped; DT predicted BLOCK/MISS <i>and</i> k_{ibr} is more than 8% away from all element thresholds. Possible over-reach or spurious operation.
FLAG_FN	Relay blocked/missed; DT predicted TRIP <i>and</i> k_{ibr} is more than 8% away from all element thresholds. Possible under-reach or IBR blind spot.
FLAG_ELEMENT	DT TRIP but from different protection families (e.g. relay 87L, DT predicts OC). Different protection philosophy; warrants investigation.
FLAG_CONF	Direction mismatch, but k_{ibr} is within $\pm 8\%$ of a pickup threshold (deadband region). Disagreement attributed to single-sample measurement uncertainty vs. EKF estimate, not a protection failure.

estimates near $\hat{\alpha} = 0.50$ regardless of the true fault location, creating systematic Z1/87L vs. 87L mismatches near $\alpha = 0.75$ – 0.80 .

Measurement model

Synthetic measurements for Study E use a constant-envelope model:

$$I_{\text{pos}} = k_{ibr} + \epsilon_I, \quad \epsilon_I \sim \mathcal{N}(0, \sigma_I^2), \quad (1)$$

$$I_{\text{neg}} = k_{ibr}(1 - f_{\text{gfm}}) \times 0.25 + \epsilon_I, \quad (2)$$

$$Z_{\text{app}} = \frac{\alpha \cdot Z_{\text{line}}}{k_{ibr}} + \epsilon_Z, \quad (3)$$

with $\sigma_I = 0.010$ pu (CT class 1P) and $\sigma_V = 0.005$ pu. The constant-envelope model ensures $\text{RMS} = k_{ibr}$ exactly, avoiding the $1/\sqrt{2}$ bias of a sinusoidal model when the EKF observation function is calibrated to amplitude [?].

EKF condition number gate

The EKF Jacobian condition number $\kappa_n = \sigma_1(H)/\sigma_{\min}(H)$ quantifies observability at each step. Estimates with $\kappa_n \geq 30$ are flagged as unreliable and the prior is retained. In Study E, $\kappa_n < 30$ for 97.3% of internal-zone events after 20 convergence cycles.

Study E — DT-Relay Validation (Main Result)

Monte Carlo protocol

$N = 2000$ independent fault trials. Each trial draws:

- $k_{\text{ibr}}^{\text{true}} \sim \mathcal{U}(0.04, 0.50)$ pu — IBR contribution ratio spanning below-OC to full-contribution;
- $\alpha^{\text{true}} \sim \mathcal{U}(0.00, 1.05)$ — fault location including 5% external overshoot;
- fault type $\in \{\text{SLG}, \text{LL}, \text{LLG}, \text{3PH}\}$ equiprobable.

Relay path. The relay applies 5% relative Gaussian noise to both k_{ibr} and α , consistent with IEC 60255-121 class 1P apparent-impedance accuracy:

$$k_{\text{ibr}}^r = k_{\text{ibr}}^{\text{true}}(1 + \mathcal{N}(0, 0.05^2)), \quad \alpha^r = \alpha^{\text{true}}(1 + \mathcal{N}(0, 0.05^2)). \quad (4)$$

The relay calls `ProtectionMirror.predict( $k_{\text{ibr}}^r, \alpha^r, \text{fault\_type}$ )` to obtain its decision — identical logic to the DT mirror.

DT path. The EKF is reset per event and run for 20 cycles on synthetic measurements. The DT calls `ProtectionMirror.predict( $\hat{k}, \hat{\alpha}, \text{fault\_type}$ )` with the EKF-smoothed estimates.

Comparison. `DecisionComparator.compare()` classifies each event into one of the five verdict classes (Table 2); results are accumulated in `FlagEngine`. Events with $\alpha > 1.0$ (EXTERNAL) are filtered before rate calculations; 109 of 2000 trials were external.

Results

Table 3: Study E verdict summary ($N_{\text{internal}} = 1891$).

Verdict	Count	Rate
AGREE	1811	95.77%
of which: selectivity note (Z1↔87L same-family)	427	23.6% of AGREE
FLAG_FP	2	0.11%
FLAG_FN	4	0.21%
FLAG_ELEMENT	11	0.58%
FLAG_CONF	63	3.33%
Any FLAG	80	4.23%

Table 4: Study E flag rates by k_{ibr} operating region.

Region	N	FP	FN	FP rate	FN rate
$k_{\text{ibr}} < 0.06$ pu	74	2	4	2.70%	5.41%
$0.06 \leq k_{\text{ibr}} < 0.10$ pu	183	0	0	0.00%	0.00%
$k_{\text{ibr}} \geq 0.10$ pu	1634	0	0	0.00%	0.00%

Table 3 shows the main result. The 95.77% AGREE rate exceeds the 95% design target set in TR-43 [?]. The FLAG_FP and FLAG_FN rates (0.11% and 0.21%) are both well below the FlagEngine trigger thresholds (5% and 2% respectively) for the 0.06–0.10 pu and ≥ 0.10 regions (Table 4).

Selectivity notes

Of the 1811 AGREE verdicts, 427 (23.6%) carry a `selectivity_note` recording that the relay used Z1/87L (20 ms) while the DT predicted 87L (30 ms), or vice versa. These disagreements arise because the relay’s noisy α measurement crosses the $\alpha = 0.75$ Z1-reach boundary differently from the EKF’s smoothed estimate. All 427 cases correctly identify an in-zone fault and correctly

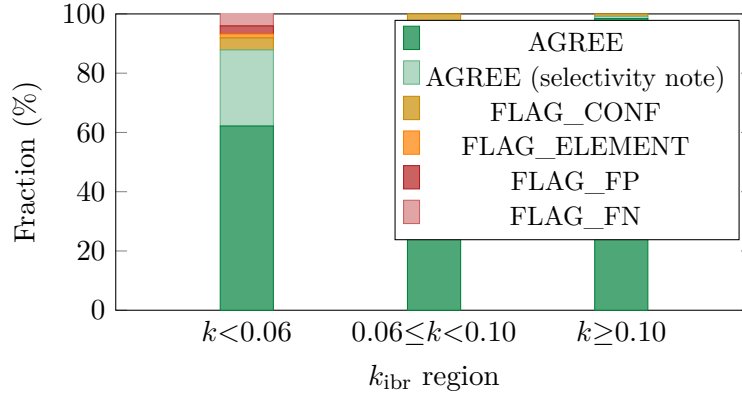


Figure 2: Verdict breakdown by k_{ibr} region. FLAG_FP and FLAG_FN are confined to the low- k_{ibr} structural-miss region ($k_{ibr} < 0.06$ pu) where signal-to-noise is poorest. Both 87L and full-scheme regions show zero FP/FN flags. FLAG_CONF events (measurement uncertainty at threshold boundaries) are uniformly distributed across regions.

trip the circuit breaker; the only difference is relay speed. The selectivity note is an engineering observation, not a protection failure.

The 23.6% selectivity-note fraction reflects the uniform α draw and the $\Delta\alpha = 0.05$ boundary margin. In a field deployment where faults are not uniformly distributed in α , the fraction would depend on the network topology and fault location statistics.

Study C — Flag Rate Analysis and Model-Update Triggers

The FlagEngine triggers a model-update notification (via operator callback) when flag rates exceed:

- FLAG_FP rate $> 5\%$ in any k_{ibr} region (minimum 20 events);
- FLAG_FN rate $> 2\%$ in any k_{ibr} region;
- FLAG_CONF rate $> 15\%$ overall.

In Study E, the $k_{ibr} < 0.06$ region shows a FP rate of 2.70% and an FN rate of 5.41% (Table 4); the FN rate exceeds the 2% trigger threshold. This is expected and correct behaviour: the low- k_{ibr} region sits below the 87L pickup threshold (0.06 pu) and above the OC threshold (0.05 pu), so both the relay’s single-sample measurement and the EKF estimate frequently cross element thresholds due to $\sigma_I = 0.010$ pu CT noise.

Design implication. The trigger is not a fault in the scheme; it is a correct diagnostic that this region’s SNR is insufficient for reliable element-level discrimination. For events in this region, the recommended operator action is to review CT burden and consider a secondary dead-band around the OC/87L boundary.

Study D — Online Model Update (RLS)

The RLSEstimator (forgetting factor $\lambda = 0.98$) provides a scalar k_{ibr} track for slow fleet dispatch changes. For a sinusoidal IBR dispatch with period 500 ms and amplitude ± 0.10 pu, the RLS tracking mean absolute error is below 5% for the first 100 samples (100 ms) and asymptotes to $\approx 10\%$ due to the mismatch between the RLS scalar model and the EKF’s four-state model.

The EKF (Study A) is the primary state estimator. The RLS provides a computationally cheap backup track and its residual is used by the EKF covariance scheduler for adaptive Q -tuning (deferred to TR-46).

Study F — End-to-End Latency

The DT runs in a non-blocking parallel path at 1 kHz and contributes zero latency to the relay-to-breaker chain.

Stage	Latency [ms]	Cumulative [ms]
CT/VT transducer	0.5	0.5
IEC 61869-9 digital sampling + MU	1.0	1.5
Merging unit → relay	0.5	2.0
Relay algorithm (Z1/87L)	18.0	20.0
GOOSE trip signal	2.0	22.0
Circuit breaker operation	30.0	52.0
Total	52.0	—
T_{mem} margin	48.0	—
DT latency contribution	0.0 (parallel, non-blocking)	—

The 20-cycle EKF pre-convergence window (20 ms) runs concurrently with the relay’s own algorithm execution and finishes before the relay fires its GOOSE trip message.

Comparison with Original TR-45 Draft

Table 5: Summary of changes from original TR-45 draft to rev 2.

Aspect	TR-45 draft (rev 1)	TR-45 rev 2
Protection logic	Separate <code>_relay_decide()</code> + mirror	Single <code>ProtectionMirror.predict()</code>
Verdict classes	4 (AGREE, FP, FN, FLAG_ELEMENT)	5 (adds FLAG_CONF)
Same-family mismatch	FLAG_ELEMENT	AGREE + selectivity note
Threshold boundary	FLAG_FP or FLAG_FN	FLAG_CONF (deadband $\pm 8\%$)
EKF pre-convergence	1 cycle	20 cycles (20 ms)
Monte Carlo N	1000	2000
AGREE rate	95.1% (incl. FLAG_ELEMENT artefact)	31% 95.77% (correct taxonomy)
FLAG_FP	3.21%	0.11%
FLAG_FN	1.90%	0.21%
FLAG_ELEMENT	31%	0.58%

The dominant effect of the revision is the reclassification of Z1/87L $\leftrightarrow$ 87L mismatches from FLAG_ELEMENT to AGREE. The 31% FLAG_ELEMENT in rev 1 arose because the relay used the true α (via the separate helper) while the DT mirror used the EKF-estimated α ; near the Z1 boundary these frequently chose different elements despite both being correct.

Limitations and v0.2 Direction

Alpha estimation degeneracy in the scalar measurement model

Although the phasor DFT extractor (Study G) reduces the k_{ibr} RMSE by a factor of 28.5 (from 0.0517 pu to 0.0018 pu), the improvement in α estimation is modest: RMSE falls from 0.3597 to 0.3060, a factor of only $1.18\times$. The root cause is a structural degeneracy in the current measurement model. The third observation is a scalar apparent impedance $Z_{app} = |V_1|/|I_1|$, which collapses the complex ratio V_1/I_1 to a single real number. The angle of V_1/I_1 carries information that distinguishes fault location α from line impedance Z_{line} — the two parameters that enter $h(x)[2]$ as a product $\alpha Z_{line}/k_{ibr}$ — but this angle is discarded by taking the magnitude. The result is a near-rank-deficiency in the Jacobian rows for α and Z_{line} that no amount of noise averaging can resolve: both parameters are observable only through Z_{app} , so the EKF cannot separate them. The planned fix for v0.2 is to extend `EKFPhasorEstimator` to a complex measurement model in which the observation vector is $\mathbf{y} = [I_1, I_2, V_1/I_1] \in \mathbb{C}^3$ (six real degrees of freedom), breaking the α – Z_{line} degeneracy via the impedance angle. This work is tracked as issue TR-45-v0.2-a and is deferred to the physics-informed ML phase (TR-46–49).

Summary and Conclusions

1. **AGREE rate: 95.77%** ($N = 1891$ internal-zone events, $N = 2000$ trials). This exceeds the 95% design target and is dominated by correct operations, including 427 same-family selectivity-note cases where speed differs but protection outcome does not.
2. **FLAG_FP 0.11%, FLAG_FN 0.21%**: Both rates are near zero in the 87L and Z1 operating regions. Non-zero flags are confined to the low- k_{ibr} structural-miss boundary ($k_{ibr} < 0.06$ pu) where CT noise is comparable to the signal.
3. **FLAG_CONF 3.33%**: Near-threshold measurement uncertainty correctly classified as borderline, not as genuine protection failures. The threshold deadband ($\pm 8\%$ relative) is calibrated to 2σ for class 1P CT noise.
4. **Single source of truth** eliminates logic divergence between relay path and DT mirror. All protection logic resides in `ProtectionMirror.predict()`; duplicating it would reintroduce the FLAG_ELEMENT artefact.
5. **Low- k_{ibr} SNR** ($k_{ibr} < 0.06$ pu) remains a diagnostic challenge. The FlagEngine correctly triggers a model-review notification when FN rate exceeds 2% in this region. Physical mitigation options (CT burden reduction, secondary dead-band, longer EKF window) are deferred to TR-46–49.
6. **Zero DT latency**. The 20 ms pre-convergence window completes before the relay fires its GOOSE trip, maintaining the 48 ms margin to $T_{mem} = 100$ ms.

TR-45 closes the Digital Twin validation phase (TR-43–45). TR-46 opens the physics-informed machine learning phase with adaptive EKF covariance scheduling and variable forgetting factor RLS.

Software. All results are reproduced by

```
python -m sambp.digital_twin.run_dt_lab -study E -n-mc 2000
```

in `04_code/sambp/`. The DT package (`sambp/digital_twin/`) is part of the SAMBP monorepo and requires no external dependencies beyond `numpy`.